



Existence and non-existence of global solutions of pseudo-parabolic equations involving $p(x)$ -Laplacian and logarithmic nonlinearity

Quach Van Chuong^{1,2,3} · Le Cong Nhan⁴ · Le Xuan Truong⁵ 

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Abstract

In this paper, we consider a class of pseudo-parabolic equations involving $p(x)$ -Laplacian and logarithmic nonlinearity. By using the potential well method, we give some threshold results on existence or nonexistence of global weak solutions when the initial energy is subcritical. Moreover, the decay estimates for the global bounded solutions and the growth estimates for the global unbounded solutions are also concerned.

Keywords Pseudo-parabolic $p(x)$ -Laplacian equation · Logarithmic nonlinearity · Blow-up · Decay.

Mathematics Subject Classification 35A01 · 35B44 · 35K55

✉ Le Xuan Truong
lxuantruong@ueh.edu.vn

Quach Van Chuong
quachuong1812@dnpu.edu.vn

Le Cong Nhan
nhanlc@hcmute.edu.vn

¹ Faculty of Mathematics and Computer Science, University of Science, Ho Chi Minh City, Vietnam

² Vietnam National University, Ho Chi Minh City, Vietnam

³ Department of Mathematics, Dong Nai University, Bien Hoa, Dong Nai, Vietnam

⁴ Faculty of Applied Sciences, Ho Chi Minh City University of Technology and Education, Ho Chi Minh City, Vietnam

⁵ Department of Mathematics and Statistics, University of Economics Ho Chi Minh City (UEH), Ho Chi Minh City, Vietnam

1 Introduction

In this paper, we study the following initial-boundary value problem:

$$\begin{cases} u_t - \Delta u_t - \Delta_{p(x)} u = |u|^{q-2} u \log |u|, & (x, t) \in \Omega \times (0, T), \\ u(x, t) = 0, & (x, t) \in \partial\Omega \times (0, T), \\ u(x, 0) = u_0(x), & x \in \Omega, \end{cases} \quad (1.1)$$

where $\Omega \subset \mathbb{R}^N$ is a bounded domain with smooth boundary, initial data

$$u_0 \in X_0 := H_0^1(\Omega) \cap W_0^{1,p(\cdot)}(\Omega),$$

$p(x)$ is a measurable function on Ω and satisfies the following *log-Hölder condition*:

$$|p(x) - p(x')| \leq \frac{C}{\log(e + 1/|x - x'|)}, \quad \text{for all } x, x' \in \Omega, \quad (1.2)$$

for some constant $C > 0$ and q is a constant in the range

$$1 < p^- \leq p^+ < q < (p^-)^* = \begin{cases} \frac{Np^-}{N - p^-}, & \text{if } p^- < N, \\ \infty, & \text{if } p^- \geq N, \end{cases} \quad (1.3)$$

where $p^- := \operatorname{ess\,inf}_{x \in \Omega} p(x)$ and $p^+ := \operatorname{ess\,sup}_{x \in \Omega} p(x)$.

It is well known that the pseudo-parabolic p -Laplacian type equations

$$u_t - \kappa \Delta u_t - \Delta_p u = f(u) \quad (1.4)$$

can be used to describe many important physical phenomena, for example, when $\kappa = 0$ the Eq. (1.4) is used in the theory of Newtonian and Non-Newtonian filtration fluids; when $\kappa > 0$ the Eq. (1.4) is used to model the flow of fluids through fissured rock, nonlinear dispersive long wave, the aggregation of populations, see [3, 4, 28].

From a mathematical point of view, the equation of the form (1.4) was first studied by Showalter [32, 33] in which the author showed the existence and uniqueness of solutions when the source term $f(u)$ is neglected. From then on, the Eq. (1.4) has attracted many researchers in a variety of different topics and one of them is related to the existence and nonexistence of global solutions to (1.4).

It is also known that the existence and nonexistence of global solutions to (1.4) depend strongly on the source type $f(u)$. In the case $f(u) = |u|^{q-2}u$ the problem (1.4) has been studied extensively (see [16, 21, 22, 34, 35] and references therein) by using potential well method introduced by Sattinger [31] (see also [29]). For the case of logarithmic nonlinearity, in [7, 8] Chen et al. considered the initial boundary problem for the equation

$$u_t - \kappa \Delta u_t - \Delta u = u \log |u|. \quad (1.5)$$

By using the potential well method introduced by Sattinger [31] (see also [29]) and the logarithmic Sobolev inequality the authors show the different effects between logarithmic nonlinearity and polynomial nonlinearity on the existence and nonexistence of global solutions. The Eq. (1.5) has been then extended to the case of pseudo-parabolic p -Laplacian equation with logarithmic nonlinearity

$$u_t - \kappa \Delta u_t - \Delta_p u = |u|^{q-2} u \log |u|,$$

where both cases of degenerate ($p \geq 2$) and singular ($1 < p < 2$) were concerned, see [5, 9, 11, 12, 19, 20, 26, 27] for example. Also using the potential well method, the following results are known:

- In [26, 27] the second and third authors considered the degenerate case $p = q > 2$ and proved the global existence and nonexistence (blow-up in finite time) of weak solutions provided that the initial energy $J(u_0)$ is less than $M \leq d$, where d is the mountain pass level. In addition, in the case global weak solutions exist, they decay algebraically. This result is then extended by He et al. [20] to the case $2 < p < q < p\left(1 + \frac{2}{N}\right)$ and $J(u_0) \leq d$. When $J(u_0) > d$ the problem was treated by Han et al. [19] for $\kappa = 0$ and by Dai et al. [9] for $\kappa > 0$.
- For the singular case $1 < p = q < 2$ the problem was studied by Cao and Liu [5], in which the authors show that the global boundedness solutions decay exponentially when $J(u_0) < M$ and in one-dimensional space. These results are then extended by Ding and Zhou in [11, 12] to the case $1 < p \leq q < p^*$.

On the other hand, in recent years much attention has been paid to the study of nonlinear parabolic equations with nonstandard growth condition that arise from modeling some physical problems such as electro-rheological fluids dynamics and image processing, etc, see [1, 2, 6, 30]. In [17], Guo et al. study the singular phenomena of solutions of the initial-boundary value problem for the equation

$$u_t - \Delta_{p(x)} u = |u|^{r-2} u.$$

The authors also point out that because of the lack of scaling property and the gap between the modular and the norm caused by the nonlinear diffusion $-\Delta_{p(x)} u$, it is not a trivial generalization of similar problems in the case of constant exponent. By using the energy method, Han [18] studied the blow-up in finite time when the initial energy is non-positive and $r > \max\{p^+, 2\}$.

Motivated by these results, we would like to study the effectiveness of logarithmic nonlinearities to the pseudo-parabolic equation involving $p(x)$ -Laplacian (1.1). It is also noticed that even in the case $p(x) \equiv p$, the decay properties of solution has not been studied (see [11, 12]). Furthermore, in the proof of Theorem 2 of [20], the authors use the Lemma 3 to prove that $\|u(t)\|_{1,p}$ decays algebraically. However, they

did not point out that $\|u(t)\|_{1,p}$ satisfies the assumption of Lemma 3, that is, $\|u(t)\|_{1,p}$ is non-increasing with respect to t . In this paper we prove this result by taking a different approach and our method can be applied to prove Theorem 2 in [20]. More precisely, we prove the decay properties of energy functional $J(u(t))$ which immediately implies decay properties of solutions to (1.1), see Theorem 4.3. In particular, in case $p^- \geq 2$ we give explicit decay estimates of global solutions by constructing a suitable Lyapunov function. So the decay properties in this paper are new even for constant exponent $p(x)$.

Notation Let $\|\cdot\|_r$ denote the usual norm of the space $L^r(\Omega)$ for $1 \leq r \leq \infty$. Moreover, $\langle \cdot, \cdot \rangle$ denote the usual inner product of the Hilbert space $L^2(\Omega)$. Finally, we denote the norm of $H_0^1(\Omega)$ by $\|\cdot\|_{H_0^1}$, that is, $\|u\|_{H_0^1} = \sqrt{\|u\|_2^2 + \|\nabla u\|_2^2}$.

2 Preliminaries results

2.1 Function spaces

This section is devoted to introduce some results on variable exponent Lebesgue spaces and variable exponent Sobolev spaces (see [10, 13–15, 23, 25] for more details).

Let Ω be an open bounded subset of $\mathbb{R}^N$ with smooth boundary. Assume that $p : \Omega \rightarrow [1, \infty]$ is a measurable function. We define the variable Lebesgue space $L^{p(\cdot)}(\Omega)$ as follows

$$L^{p(\cdot)}(\Omega) := \left\{ u : \Omega \rightarrow \mathbb{R} \text{ is measurable, } \rho(u) := \int_{\Omega} |u(x)|^{p(x)} dx < \infty \right\}.$$

This space is equipped with the Luxemburg-type norm

$$\|u\|_{p(\cdot)} := \inf \left\{ \lambda > 0 : \int_{\Omega} \left| \frac{u(x)}{\lambda} \right|^{p(x)} dx \leq 1 \right\}.$$

The following proposition shows the relation between the norm $\|u\|_{p(\cdot)}$ and the modular $\rho(u)$.

Proposition 1 [10] *Let $p \in \mathcal{P}(\Omega)$. It holds that*

$$\min \left\{ \|u\|_{p(\cdot)}^{p^-}, \|u\|_{p(\cdot)}^{p^+} \right\} \leq \rho(u) \leq \max \left\{ \|u\|_{p(\cdot)}^{p^-}, \|u\|_{p(\cdot)}^{p^+} \right\}, \quad \text{for all } u \in L^{p(\cdot)}(\Omega).$$

We also note that, for $p^+ < \infty$, the dual space of $L^{p(\cdot)}(\Omega)$ is identified with $L^{p'(\cdot)}(\Omega)$ where

$$\frac{1}{p(x)} + \frac{1}{p'(x)} = 1 \quad \text{for a.e. } x \in \Omega.$$

Proposition 2 (Hölder inequality, [10]) *Let p, q, s are measurable functions functions on Ω with values $[1, \infty]$. Assume that*

$$\frac{1}{s(x)} = \frac{1}{p(x)} + \frac{1}{q(x)} \quad \text{for a.e. } x \in \Omega.$$

Then we have

$$\|uv\|_{s(\cdot)} \leq 2\|u\|_{p(\cdot)}\|v\|_{q(\cdot)} \quad \text{for all } u \in L^{p(\cdot)}(\Omega), v \in L^{q(\cdot)}(\Omega),$$

Proposition 3 [10] *Let p, q are measurable functions functions on Ω with values $[1, \infty]$. If $q(x) \geq p(x)$ for a.e. $x \in \Omega$ then the embedding $L^{q(\cdot)}(\Omega) \hookrightarrow L^{p(\cdot)}(\Omega)$ is continuous.*

We next define variable exponent Sobolev spaces $W^{1,p(\cdot)}(\Omega)$ as follows:

$$W^{1,p(\cdot)}(\Omega) = \{u \in L^{p(\cdot)}(\Omega) : |\nabla u| \in L^{p(\cdot)}(\Omega)\},$$

endowed with the norm

$$\|u\|_{W^{1,p(\cdot)}(\Omega)} = \left(\|u\|_{p(\cdot)}^2 + \|\nabla u\|_{p(\cdot)}^2 \right)^{1/2}.$$

Moreover we also define $W_0^{1,p(\cdot)}(\Omega)$ as the closure of $C_0^\infty(\Omega)$ in $W^{1,p(\cdot)}(\Omega)$. It is noticed that if

$$1 < p^- \leq p^+ < \infty$$

then $L^{p(\cdot)}(\Omega)$ and $W^{1,p(\cdot)}(\Omega)$ are uniformly convex Banach spaces and therefore they are reflexive.

To end this section we will state the Poincaré and Sobolev inequalities. To do this we need condition (1.2), which follows from a Hölder continuity of p over Ω with any Hölder exponent and it implies $p \in C(\overline{\Omega})$ and $p^+ < \infty$.

Proposition 4 [10] *Let $p : \Omega \rightarrow [1, \infty]$ be a measurable function. If the condition (1.2) holds, then*

(i) *There exists a constant $C \geq 0$ such that*

$$\|u\|_{p(\cdot)} \leq C\|\nabla u\|_{p(\cdot)} \quad \text{for all } u \in W_0^{1,p(\cdot)}(\Omega).$$

In particular, the space $W_0^{1,p(\cdot)}(\Omega)$ possesses an equivalent norm $\|\cdot\|_{1,p(\cdot)}$ given by

$$\|u\|_{1,p(\cdot)} := \|\nabla u\|_{p(\cdot)} \quad \text{for } u \in W_0^{1,p(\cdot)}(\Omega).$$

(ii) Let $q : \Omega \rightarrow [1, \infty)$ be a measurable and bounded function and suppose that

$$q(x) \leq p^*(x) := \begin{cases} \frac{Np(x)}{N-p(x)}, & \text{if } p(x) < N, \\ \infty, & \text{if } p(x) \geq N, \end{cases}$$

for a.e. $x \in \Omega$. Then $W^{1,p(\cdot)}(\Omega)$ is continuously embedded in $L^{q(\cdot)}(\Omega)$. In addition, assume that

$$\operatorname{ess\,inf}_{x \in \Omega} (p^*(x) - q(x)) > 0.$$

Then the embedding $W_0^{1,p(\cdot)}(\Omega) \hookrightarrow L^{q(\cdot)}(\Omega)$ is compact.

2.2 Stationary problem and Potential wells

In this section, we will study the stationary solutions of (1.1) which solve the problem

$$\begin{cases} -\Delta_{p(x)} u = |u|^{q-2} u \log |u|, & \text{in } \Omega, \\ u(x) = 0, & \text{on } \partial\Omega, \end{cases} \quad (2.1)$$

where $p(x)$ and q satisfies (1.2)–(1.3).

Consider the energy functional J and the Nehari functional I given by

$$\begin{aligned} J(u) &= \int_{\Omega} \frac{1}{p(x)} |\nabla u|^{p(x)} \, dx - \frac{1}{q} \int_{\Omega} |u|^q \log |u| \, dx + \frac{1}{q^2} \|u\|_q^q, \\ I(u) &= \int_{\Omega} |\nabla u|^{p(x)} \, dx - \int_{\Omega} |u|^q \log |u| \, dx. \end{aligned}$$

Then I and J belong to the class C^1 over $W_0^{1,p(\cdot)}(\Omega)$. The critical points of J are also weak solutions of (2.1). Furthermore, it is clear that

$$J(u) = \int_{\Omega} \left(\frac{1}{p(x)} - \frac{1}{q} \right) |\nabla u|^{p(x)} \, dx + \frac{1}{q^2} \|u\|_q^q + \frac{1}{q} I(u). \quad (2.2)$$

Let $u \in W_0^{1,p(\cdot)}(\Omega) \setminus \{0\}$ and consider the fibering map $s \mapsto j(s)$ given by

$$\begin{aligned} j(s) &:= J(su) = \int_{\Omega} \frac{s^{p(x)}}{p(x)} |\nabla u|^{p(x)} \, dx - \frac{s^q}{q} \int_{\Omega} |u|^q \log |u| \, dx \\ &\quad + \frac{s^q}{q^2} \|u\|_q^q - \frac{s^q \log s}{q} \|u\|_q^q. \end{aligned}$$

We have the following lemma.

Lemma 2.1 *Let (1.2)–(1.3) hold and $u \in W_0^{1,p(\cdot)}(\Omega) \setminus \{0\}$. Then the following results hold:*

- (i) $\lim_{s \rightarrow 0^+} j(s) = 0$ and $\lim_{s \rightarrow \infty} j(s) = -\infty$.
- (ii) *There exists a positive number $s_* = s_*(u)$ such that $j(s)$ attains the maximum at $s = s_*$, then $I(s_*u) = 0$. In addition, we have $0 < s_* < 1$, $s_* = 1$ and $s_* > 1$ provided that $I(u) < 0$, $I(u) = 0$ and $I(u) > 0$, respectively.*

Proof It is easily seen that

$$\begin{aligned} j(s) &\geq \min \{s^{p^-}, s^{p^+}\} \int_{\Omega} \frac{1}{p(x)} |\nabla u|^{p(x)} dx - \frac{s^q}{q} \int_{\Omega} |u|^q \log |u| dx \\ &\quad + \frac{s^q}{q^2} \|u\|_q^q - \frac{s^q \log s}{q} \|u\|_q^q, \end{aligned}$$

and

$$\begin{aligned} j(s) &\leq \max \{s^{p^-}, s^{p^+}\} \int_{\Omega} \frac{1}{p(x)} |\nabla u|^{p(x)} dx - \frac{s^q}{q} \int_{\Omega} |u|^q \log |u| dx \\ &\quad + \frac{s^q}{q^2} \|u\|_q^q - \frac{s^q \log s}{q} \|u\|_q^q. \end{aligned}$$

This, together with $q > p^+$ and $\|u\|_q^q > 0$, implies (i). Furthermore, we also have $j(s) > 0$ for sufficiently small $s > 0$. Hence, there exists a $s_* > 0$ such that $j(s_*) = \sup_{s>0} j(s)$. Then by Fermat's Theorem, we obtain $j'(s_*) = 0$. This implies that $I(s_*u) = 0$ by the relation $I(su) = sj'(s)$.

Finally, we prove the last statement of (ii). By the definition of I , we get

$$\begin{aligned} 0 = I(s_*u) &= \int_{\Omega} s_*^{p(x)} |\nabla u|^{p(x)} dx - s_*^q \int_{\Omega} |u|^q \log |u| dx - s_*^q \log s_* \|u\|_q^q \\ &= \int_{\Omega} s_*^{p(x)} |\nabla u|^{p(x)} dx - s_*^q \left(\int_{\Omega} |\nabla u|^{p(x)} dx - I(u) \right) - s_*^q \log s_* \|u\|_q^q \\ &= \int_{\Omega} (s_*^{p(x)} - s_*^q) |\nabla u|^{p(x)} dx - s_*^q \log s_* \|u\|_q^q + s_*^q I(u), \end{aligned}$$

which can be rewritten as

$$s_*^q I(u) = \int_{\Omega} (s_*^q - s_*^{p(x)}) |\nabla u|^{p(x)} dx + s_*^q \log s_* \|u\|_q^q.$$

Since $q > p^+$, the above equality shows that $0 < s_* < 1$, $s_* = 1$ and $s_* > 1$ provided that $I(u) < 0$, $I(u) = 0$ and $I(u) > 0$, respectively. The proof is complete. $\square$

We now define the so-called Nehari manifold associated to the energy functional J by

$$\mathcal{N} = \left\{ u \in W_0^{1,p(\cdot)}(\Omega) \setminus \{0\} : I(u) = 0 \right\}.$$

It follows from Lemma 2.1 that $\mathcal{N}$ is not empty set. Hence we can define

$$d = \inf_{u \in \mathcal{N}} J(u).$$

We also define the so-called stable set $\mathcal{W}$ and unstable set $\mathcal{U}$ which is similar to Sattinger [31], Payne and Sattinger [29].

$$\begin{aligned} \mathcal{W} &= \left\{ u \in W_0^{1,p(\cdot)}(\Omega) : J(u) < d, I(u) > 0 \right\} \cup \{0\}, \\ \mathcal{U} &= \left\{ u \in W_0^{1,p(\cdot)}(\Omega) : J(u) < d, I(u) < 0 \right\}. \end{aligned}$$

Through simple calculations, we have the following lemma. So we omit the proof here.

Lemma 2.2 *Let $a > 0$. Then the following inequality holds:*

$$\log x \leq \frac{x^a}{ea}, \quad \text{for all } x > 0.$$

We next prove the following lemma, which shows that d is a positive number and is actually attained at some $u \in \mathcal{N}$.

Lemma 2.3 *Let (1.2)–(1.3) hold. Then we have*

- (i) $d = \inf_{u \in W_0^{1,p(\cdot)}(\Omega) \setminus \{0\}} \sup_{s>0} J(su).$
- (ii) d is a positive number.
- (iii) For any $u \in W_0^{1,p(\cdot)}(\Omega)$. If $I(u) < 0$, then $J(u) - \frac{1}{q}I(u) > d$.
- (iv) There exists $u^* \in \mathcal{N}$ such that $J(u^*) = d$.

Proof We first prove (i). Let any $u \in W_0^{1,p(\cdot)}(\Omega) \setminus \{0\}$. By virtue of Lemma 2.1, we have

$$\sup_{s>0} J(su) = J(s_*u). \quad (2.3)$$

By the definition of $\mathcal{N}$, it follows from Lemma 2.1 that $s_*u \in \mathcal{N}$. Hence,

$$J(s_*u) \geq \inf_{u \in \mathcal{N}} J(u) = d. \quad (2.4)$$

Combining (2.3) and (2.4), one has

$$\inf_{u \in W_0^{1,p(\cdot)}(\Omega) \setminus \{0\}} \sup_{s>0} J(su) \geq d. \quad (2.5)$$

On the other hand, for any $u \in \mathcal{N}$, by virtue of Lemma 2.1, we get $s_* = 1$, that is,

$$\sup_{s>0} J(su) = J(u).$$

Hence,

$$\inf_{u \in W_0^{1,p(\cdot)}(\Omega) \setminus \{0\}} \sup_{s>0} J(su) \leq \inf_{u \in \mathcal{N}} \sup_{s>0} J(su) = \inf_{u \in \mathcal{N}} J(u) = d. \quad (2.6)$$

We deduce from (2.5) and (2.6) that (i) holds true.

In order to prove (ii), we choose $\epsilon > 0$ small enough so that

$$q + \epsilon < (p^-)^*. \quad (2.7)$$

Let any $u \in W_0^{1,p(\cdot)}(\Omega) \setminus \{0\}$ such that $I(u) \leq 0$. Then by using Lemma 2.2, we get

$$\begin{aligned} \min \left\{ \|\nabla u\|_{p(\cdot)}^{p^-}, \|\nabla u\|_{p(\cdot)}^{p^+} \right\} &\leq \int_{\Omega} |\nabla u|^{p(x)} \, dx \leq \int_{\Omega} |u|^q \log |u| \, dx \\ &\leq \frac{1}{e\epsilon} \|u\|_{q+\epsilon}^{q+\epsilon} \\ &\leq \frac{1}{e\epsilon} S_{q+\epsilon}^{q+\epsilon} \|\nabla u\|_{p(\cdot)}^{q+\epsilon}, \end{aligned}$$

where $S_{q+\epsilon}$ is the best imbedding constant of $W_0^{1,p(\cdot)}(\Omega) \hookrightarrow L^{q+\epsilon}(\Omega)$. Hence, since $\|\nabla u\|_{p(\cdot)} > 0$ and $q > p^+$, we obtain

$$\|\nabla u\|_{p(\cdot)} \geq \delta_1, \quad (2.8)$$

where

$$\delta_1 = \min \left\{ (e\epsilon)^{\frac{1}{q+\epsilon-p^-}} S_{q+\epsilon}^{-\frac{q+\epsilon}{q+\epsilon-p^-}}, (e\epsilon)^{\frac{1}{q+\epsilon-p^+}} S_{q+\epsilon}^{-\frac{q+\epsilon}{q+\epsilon-p^+}} \right\} > 0.$$

For any $u \in \mathcal{N}$, it follows from (2.2) and (2.8) that

$$\begin{aligned} J(u) &= \int_{\Omega} \left(\frac{1}{p(x)} - \frac{1}{q} \right) |\nabla u|^{p(x)} \, dx + \frac{1}{q^2} \|u\|_q^q + \frac{1}{q} I(u) \\ &\geq \left(\frac{1}{p^+} - \frac{1}{q} \right) \min \left\{ \|\nabla u\|_{p(\cdot)}^{p^-}, \|\nabla u\|_{p(\cdot)}^{p^+} \right\} \\ &\geq \left(\frac{1}{p^+} - \frac{1}{q} \right) \min \left\{ \delta_1^{p^-}, \delta_1^{p^+} \right\}. \end{aligned} \quad (2.9)$$

Then by the definition of d , we imply that

$$d \geq \left(\frac{1}{p^+} - \frac{1}{q} \right) \min \left\{ \delta_1^{p^-}, \delta_1^{p^+} \right\} > 0.$$

Now we shall prove (iii). Let $u \in W_0^{1,p(\cdot)}(\Omega)$ satisfying $I(u) < 0$. Then by virtue of Lemma 2.1, there exists $s_* \in (0, 1)$ such that $I(s_*u) = 0$. By the definition of d and replacing u by s_*u in the equality (2.2), one has

$$\begin{aligned} d \leq J(s_*u) &= \int_{\Omega} s_*^{p(x)} \left(\frac{1}{p(x)} - \frac{1}{q} \right) |\nabla u|^{p(x)} dx + \frac{s_*^q}{q^2} \|u\|_q^q \\ &< \int_{\Omega} \left(\frac{1}{p(x)} - \frac{1}{q} \right) |\nabla u|^{p(x)} dx + \frac{1}{q^2} \|u\|_q^q \\ &= J(u) - \frac{1}{q} I(u). \end{aligned}$$

Finally, we prove (iv).

By the definition of d , there exists $\{u_n\}_{n=1}^{\infty} \subset \mathcal{N}$ being a minimizing sequence of J such that $\lim_{n \rightarrow \infty} J(u_n) = d$. From this and using (2.9), we get $\{u_n\}$ is bounded in $W_0^{1,p(\cdot)}(\Omega)$. Then since $W_0^{1,p(\cdot)}(\Omega)$ is reflexive, $W_0^{1,p(\cdot)}(\Omega) \hookrightarrow L^{q+\epsilon}(\Omega)$ is compact embedding with $\epsilon > 0$ satisfying (2.7), there exists a subsequence of $\{u_n\}$, still denoted by $\{u_n\}$ and a $u^* \in W_0^{1,p(\cdot)}(\Omega)$ such that

$$\begin{aligned} u_n &\rightharpoonup u^* \quad \text{weakly in } W_0^{1,p(\cdot)}(\Omega), \\ u_n &\rightarrow u^* \quad \text{strongly in } L^{q+\epsilon}(\Omega), \\ u_n(x) &\rightarrow u^*(x) \quad \text{a.e. in } \Omega. \end{aligned}$$

Hence, we can deduce

$$\begin{aligned} \int_{\Omega} |\nabla u^*|^{p(x)} dx &\leq \liminf_{n \rightarrow \infty} \int_{\Omega} |\nabla u_n|^{p(x)} dx, \\ \int_{\Omega} \left(\frac{1}{p(x)} - \frac{1}{q} \right) |\nabla u^*|^{p(x)} dx &\leq \liminf_{n \rightarrow \infty} \int_{\Omega} \left(\frac{1}{p(x)} - \frac{1}{q} \right) |\nabla u_n|^{p(x)} dx, \\ \|u^*\|_q &= \lim_{n \rightarrow \infty} \|u_n\|_q, \\ \int_{\Omega} |u^*|^q \log |u^*| dx &= \lim_{n \rightarrow \infty} \int_{\Omega} |u_n|^q \log |u_n| dx. \end{aligned}$$

Because $u_n \in \mathcal{N}$, one has

$$\begin{aligned}
d = \liminf_{n \rightarrow \infty} J(u_n) &= \liminf_{n \rightarrow \infty} \left[\int_{\Omega} \left(\frac{1}{p(x)} - \frac{1}{q} \right) |\nabla u_n|^{p(x)} dx + \frac{1}{q^2} \|u_n\|_q^q \right] \\
&\geq \int_{\Omega} \left(\frac{1}{p(x)} - \frac{1}{q} \right) |\nabla u^*|^{p(x)} dx + \frac{1}{q^2} \|u^*\|_q^q \\
&= J(u^*) - \frac{1}{q} I(u^*).
\end{aligned} \tag{2.10}$$

Then by virtue of Lemma 2.3 (iii), this leads to

$$I(u^*) \geq 0. \tag{2.11}$$

On the other hand, we also have

$$\begin{aligned}
0 &= \liminf_{n \rightarrow \infty} I(u_n) = \liminf_{n \rightarrow \infty} \left(\int_{\Omega} |\nabla u_n|^{p(x)} dx - \int_{\Omega} |u_n|^q \log |u_n| dx \right) \\
&\geq \int_{\Omega} |\nabla u^*|^{p(x)} dx - \int_{\Omega} |u^*|^q \log |u^*| dx \\
&= I(u^*).
\end{aligned}$$

This together with (2.11), implies that $I(u^*) = 0$. To prove that $u^* \in \mathcal{N}$, it remains to check $u^* \neq 0$. Indeed, since $u_n \in \mathcal{N}$ and (2.8), we obtain

$$\begin{aligned}
\int_{\Omega} |u_n|^q \log |u_n| dx &= \int_{\Omega} |\nabla u_n|^{p(x)} dx \geq \min \left\{ \|\nabla u_n\|_{p(\cdot)}^{p^-}, \|\nabla u_n\|_{p(\cdot)}^{p^+} \right\} \\
&\geq \min \left\{ \delta_1^{p^-}, \delta_1^{p^+} \right\}.
\end{aligned}$$

Letting $n \rightarrow \infty$, we have

$$\int_{\Omega} |u^*|^q \log |u^*| dx \geq \min \left\{ \delta_1^{p^-}, \delta_1^{p^+} \right\} > 0,$$

which implies that $u^* \neq 0$. Since $u^* \in \mathcal{N}$, we have $J(u^*) \geq d$. Moreover, by (2.10) and $I(u^*) = 0$, it is clear that $J(u^*) \leq d$. Hence $J(u^*) = d$. The proof is complete. $\square$

3 Nonexistence of global solutions

We start this section by providing the precise meaning of solution to problem (1.1).

Definition 3.1 (*Weak solution*) Let $T > 0$ be an arbitrary constant and $u_0 \in X_0$. A function

$$u \in L^\infty(0, T; X_0) \text{ with } u_t \in L^2(0, T; H_0^1(\Omega))$$

is said to be a *weak solution* to (1.1) in $\Omega \times [0, T)$ if $u(0) = u_0$ and the following variational formula holds for any $v \in X_0$

$$\langle u_t, v \rangle + \langle \nabla u_t, \nabla v \rangle + \langle |\nabla u|^{p(x)-2} \nabla u, \nabla v \rangle = \langle |u|^{q-2} u \log |u|, v \rangle, \quad \text{a.e. } t \in (0, T).$$

The local existence of weak solutions to problem (1.1) is given in following theorem.

Theorem 3.2 *Let the assumptions (1.2)–(1.3) hold. Then for any initial function $u_0 \in X_0$, there exists a positive constant T_0 such that the problem (1.1) has a weak solution $u(x, t)$ on $\Omega \times (0, T_0)$. Moreover, it also satisfies the energy inequality*

$$\int_0^t \|u'(s)\|_{H_0^1}^2 ds + J(u(t)) \leq J(u_0), \quad 0 \leq t < T_0.$$

Proof The proof of this theorem is based on the standard Faedo–Galerkin’s approximation technique and associated with the weak compact arguments (see [24] or [27] for further details). So we omit its proof here. $\square$

Definition 3.3 (*Maximal existence time*) Let $u(t)$ be a weak solution to the problem (1.1). We define the maximal existence time $T_{\max}$ of $u(t)$ by

- (i) $T_{\max} = \infty$ if $u(t)$ exists for $0 \leq t < \infty$,
- (ii) $T_{\max} = T_0 > 0$ if $u(t)$ exists for $0 \leq t < T_0$, but does not exist at T_0 .

Now we concern the nonexistence of global weak solutions to problem (1.1) in the both case

$$J(u_0) \leq 0,$$

or

$$J(u_0) < d \quad \text{and} \quad I(u_0) < 0.$$

Moreover, we also obtain the upper bounds for the maximal existence time. More precisely, in the first case we have the following theorem:

Theorem 3.4 *Let (1.2)–(1.3) hold and $p^- > 2$. Assume that the nonzero initial datum $u_0 \in X_0$ satisfying $J(u_0) \leq 0$. Then the problem (1.1) has no global weak solution. Furthermore, the maximal existence time possesses an upper bound as follows*

$$T_{\max} \leq C \max \left\{ \|u_0\|_{H_0^1}^{2-p^-}, \|u_0\|_{H_0^1}^{2-p^+} \right\},$$

where the constant C defined by

$$C = \frac{p^- \max \{S_H^{p^-}, S_H^{p^+}\}}{(p^- - 2)(q - p^+)} > 0, \quad (3.1)$$

with S_H is the optimal embedding constant of $W_0^{1,p(\cdot)}(\Omega) \hookrightarrow H_0^1(\Omega)$ which means that

$$S_H = \sup_{u \in W_0^{1,p(\cdot)}(\Omega) \setminus \{0\}} \frac{\|u\|_{H_0^1}}{\|\nabla u\|_{p(\cdot)}}. \quad (3.2)$$

Proof Let u be a weak solution to problem (1.1) on $[0, T_{\max})$. We define the function $f(t)$ by

$$f(t) = \|u(t)\|_{H_0^1}^2, \quad \text{for all } t \in [0, T_{\max}).$$

By the definitions of I, J and the fact $J(u(t)) \leq J(u_0) \leq 0$ it follows that

$$\begin{aligned} f'(t) &= 2 \int_{\Omega} u'(t)u(t) \, dx + 2 \int_{\Omega} \nabla u'(t) \nabla u(t) \, dx = -2I(u(t)) \\ &= -2qJ(u(t)) + 2q \int_{\Omega} \left(\frac{1}{p(x)} - \frac{1}{q} \right) |\nabla u(t)|^{p(x)} \, dx + \frac{2}{q} \|u(t)\|_q^q \\ &\geq 2q \left(\frac{1}{p^+} - \frac{1}{q} \right) \int_{\Omega} |\nabla u(t)|^{p(x)} \, dx. \end{aligned} \quad (3.3)$$

This implies that f is increasing on $[0, T_{\max})$. Hence we imply

$$f(t) \geq f(0) = \|u_0\|_{H_0^1}^2 > 0, \quad \text{for all } t \in [0, T_{\max}).$$

So, it follows from (3.3) that

$$\begin{aligned} f'(t) &\geq 2q \left(\frac{1}{p^+} - \frac{1}{q} \right) \min \left\{ \|\nabla u(t)\|_{p(\cdot)}^{p^-}, \|\nabla u(t)\|_{p(\cdot)}^{p^+} \right\} \\ &\geq 2q \left(\frac{1}{p^+} - \frac{1}{q} \right) \min \left\{ S_H^{-p^-} \|u(t)\|_{H_0^1}^{p^-}, S_H^{-p^+} \|u(t)\|_{H_0^1}^{p^+} \right\} \\ &\geq 2q \left(\frac{1}{p^+} - \frac{1}{q} \right) \min \left\{ S_H^{-p^-}, S_H^{-p^+} \right\} \min \left\{ \|u(t)\|_{H_0^1}^{p^-}, \|u(t)\|_{H_0^1}^{p^+} \right\} \\ &= 2q \left(\frac{1}{p^+} - \frac{1}{q} \right) \min \left\{ S_H^{-p^-}, S_H^{-p^+} \right\} \min \left\{ 1, f^{\frac{p^+ - p^-}{2}}(t) \right\} f^{\frac{p^-}{2}}(t) \\ &\geq 2q \left(\frac{1}{p^+} - \frac{1}{q} \right) \min \left\{ S_H^{-p^-}, S_H^{-p^+} \right\} \min \left\{ 1, \|u_0\|_{H_0^1}^{p^+ - p^-} \right\} f^{\frac{p^-}{2}}(t). \end{aligned} \quad (3.4)$$

Since $f(t) > 0$, the inequality (3.4) can be rewritten as

$$f'(t)f^{-\frac{p^-}{2}}(t) \geq C_0, \quad (3.5)$$

where

$$C_0 = 2q \left(\frac{1}{p^+} - \frac{1}{q} \right) \min \left\{ S_H^{-p^-}, S_H^{-p^+} \right\} \min \left\{ 1, \|u_0\|_{H_0^1}^{p^+-p^-} \right\} > 0.$$

Integrating (3.5) over $[0, t]$, one has

$$f^{\frac{2-p^-}{2}}(t) \leq f^{\frac{2-p^-}{2}}(0) - \left(\frac{p^-}{2} - 1 \right) C_0 t, \quad \text{for all } t \in [0, T_{\max}),$$

which, together with $f^{\frac{2-p^-}{2}}(t) > 0$, implies

$$t < \frac{2}{(p^- - 2)C_0} \|u_0\|_{H_0^1}^{2-p^-}, \quad \text{for all } t \in [0, T_{\max}).$$

Hence,

$$T_{\max} \leq \frac{2}{(p^- - 2)C_0} \|u_0\|_{H_0^1}^{2-p^-} = C \max \left\{ \|u_0\|_{H_0^1}^{2-p^-}, \|u_0\|_{H_0^1}^{2-p^+} \right\},$$

where C is defined in (3.1). The proof is complete. $\square$

Next the result of nonexistence of global weak solutions to problem (1.1) in the second case is stated in the following theorem.

Theorem 3.5 *Let (1.2)–(1.3) hold and $q > 2$. Assume further that the initial datum u_0 belongs to the unstable set $\mathcal{U}$. Then there is no global solution to problem (1.1). Moreover, the maximal existence time $T_{\max} < \infty$ possesses an upper bound, namely we have*

$$T_{\max} \leq \frac{4(q-1)\|u_0\|_2^2}{q(q-2)^2(d - J(u_0))}.$$

Proof Define the function F by

$$F(t) = \int_0^t \|u(s)\|_{H_0^1}^2 ds, \quad t \in [0, T_{\max}). \quad (3.6)$$

Then we have that

$$\begin{aligned} F''(t) &= 2 \int_{\Omega} u'(t)u(t) dx + 2 \int_{\Omega} \nabla u'(t) \nabla u(t) dx = -2I(u(t)), \\ F'(t) &= \|u(t)\|_{H_0^1}^2 = 2 \int_0^t \left[\int_{\Omega} u'(s)u(s) dx + \int_{\Omega} \nabla u'(s) \nabla u(s) dx \right] ds + \|u_0\|_{H_0^1}^2. \end{aligned}$$

We start by proving that $I(u(t)) < 0$ for all $0 \leq t < T_{\max}$. Indeed, assume on the contrary that there exists $t_0 < T_{\max}$ such that

$$I(u(t)) < 0, \quad \forall t \in [0, t_0) \quad \text{and} \quad I(u(t_0)) = 0.$$

Then since $J(u(t_0)) \leq J(u_0) < d$, we get $u(t_0) \notin \mathcal{N}$ and hence $u(t_0) = 0$. On the other hand, since (2.8), we get $\|\nabla u(t)\|_{p(\cdot)} \geq \delta_1$, for all $t \in [0, t_0)$. Letting $t \rightarrow t_0$, we obtain $\|\nabla u(t_0)\|_{p(\cdot)} \geq \delta_1 > 0$, which gives $u(t_0) \neq 0$. We thus arrive at a contradiction.

Then by virtue of Lemma 2.3 (iii), we get

$$J(u(t)) - \frac{1}{q}I(u(t)) > d,$$

which implies that

$$F''(t) = -2I(u(t)) > 2q(d - J(u(t))) = 2q(d - J(u_0)) + 2q \int_0^t \|u'(s)\|_{H_0^1}^2 ds. \quad (3.7)$$

It is from this that

$$\begin{aligned} F'(t) &= F'(0) + \int_0^t F''(s) ds \\ &\geq \|u_0\|_{H_0^1}^2 + 2q(d - J(u_0))t > 0, \end{aligned} \quad (3.8)$$

which yields

$$F(t) > F(0) = 0, \quad \text{for all } t > 0.$$

Let $T \in (0, T_{\max})$ and consider the function $K(t)$ defined by

$$K(t) = F(t) + (T - t)F'(0) + a(t + b)^2, \quad \text{for } t \in [0, T].$$

Here a and b are two positive constants given later. We then have

$$\begin{aligned} K'(t) &= F'(t) - F'(0) + 2a(t + b) > 0, \\ K''(t) &= F''(t) + 2a > 0. \end{aligned}$$

Using Cauchy–Schwarz inequality, one has

$$\begin{aligned}
K(t) \left(\int_0^t \|u'(s)\|_{H_0^1}^2 ds + a \right) &\geq \left(\int_0^t \|u(s)\|_{H_0^1}^2 ds + a(t+b)^2 \right) \left(\int_0^t \|u'(s)\|_{H_0^1}^2 ds + a \right) \\
&\geq \left(\sqrt{\int_0^t \|u'(s)\|_{H_0^1}^2 ds} \int_0^t \|u(s)\|_{H_0^1}^2 ds + a(t+b) \right)^2 \\
&\geq \left(\int_0^t \left[\int_{\Omega} u'(s)u(s) dx + \int_{\Omega} \nabla u'(s) \nabla u(s) dx \right] ds + a(t+b) \right)^2 \\
&= \frac{1}{4} (F'(t) - F'(0) + 2a(t+b))^2 \\
&= \frac{1}{4} (K'(t))^2.
\end{aligned}$$

From this and (3.7), we get

$$\begin{aligned}
K''(t)K(t) - \frac{q}{2} (K'(t))^2 &\geq K(t) \left(K''(t) - 2q \left(\int_0^t \|u'(s)\|_2^2 ds + a \right) \right) \\
&= K(t) \left(F''(t) - 2q \int_0^t \|u'(s)\|_2^2 ds + 2a(1-q) \right) \\
&\geq K(t) (2q(d - J(u_0)) + 2a(1-q)).
\end{aligned}$$

Choosing $a = \frac{q(d - J(u_0))}{q-1} > 0$, we have

$$K''(t)K(t) - \frac{q}{2} (K'(t))^2 \geq 0.$$

Setting $G(t) = K^{1-\frac{q}{2}}(t)$, we get

$$G'(t) = \left(1 - \frac{q}{2}\right) \frac{K'(t)}{K^{\frac{q}{2}}(t)}, \quad \text{and} \quad G''(t) = \left(1 - \frac{q}{2}\right) \frac{K(t)K''(t) - \frac{q}{2}(K'(t))^2}{K^{1+\frac{q}{2}}(t)}.$$

Then we have $G''(t) \leq 0$, for all $t \in [0, T]$. Thus, $G(t)$ is concave on $[0, T]$. This implies that

$$G(T) \leq G(0) + TG'(0).$$

From this and notice that $G(T) > 0$ and $G'(0) < 0$, we get

$$T \leq -\frac{G(0)}{G'(0)} = \frac{K(0)}{\left(\frac{q}{2} - 1\right)K'(0)} = \frac{T\|u_0\|_2^2 + ab^2}{(q-2)ab}.$$

By choosing $b > \frac{\|u_0\|_2^2}{(q-2)a}$, we get

$$T \leq \frac{ab^2}{(q-2)ab - \|u_0\|_2^2} := f(b).$$

By simple calculations, one has

$$\min_{b > \frac{\|u_0\|_2^2}{(q-2)a}} f(b) = f\left(\frac{2\|u_0\|_2^2}{(q-2)a}\right) = \frac{4\|u_0\|_2^2}{(q-2)^2 a} = \frac{4(q-1)\|u_0\|_2^2}{q(q-2)^2(d - J(u_0))}.$$

Then since $T \in (0, T_{\max})$ is an arbitrary constant, we imply that

$$T_{\max} \leq \frac{4(q-1)\|u_0\|_2^2}{q(q-2)^2(d - J(u_0))}.$$

The proof is complete. $\square$

4 Global existence and asymptotic behavior

In this section, we obtain the global existence results of weak solutions. Moreover, the large time asymptotic behavior of solutions are also provided.

Theorem 4.1 *Let (1.2)–(1.3) hold. If $u_0 \in X_0$ then the weak solutions of problem (1.1) are global provided that one of two following cases occurs*

- (a) $q < 2$, or
- (b) $u_0 \in \mathcal{W}$.

Proof We shall use the Faedo–Galerkin approximation method. Let $\{w_j\}_{j=1}^\infty$ be a system of basic functions in $X_0 = H_0^1(\Omega) \cap W_0^{1,p(\cdot)}(\Omega)$. We construct the approximate solution $u^k(x, t)$ of the problem (1.1) by the form

$$u^k(x, t) = \sum_{j=1}^k g_j(t) w_j(x), \quad k = 1, 2, \dots,$$

where the coefficients $g_j(t)$ satisfy the system of ordinary differential equations

$$\langle u_t^k, w_j \rangle + \langle \nabla u_t^k, \nabla w_j \rangle + \left\langle | \nabla u^k |^{p(x)-2} \nabla u^k, \nabla w_j \right\rangle = \left\langle | u^k |^{q-2} u^k \log | u^k |, w_j \right\rangle, \quad (4.9)$$

$$u^k(0) = \sum_{j=1}^k g_j(0)w_j(x) \rightarrow u_0 \quad \text{in } X_0 \quad \text{as } k \rightarrow \infty, \quad (4.10)$$

for $j = 1, 2, \dots, k$. The existence of a local solution of the above system is guaranteed by Peano's theorem.

Multiplying (4.9) by $g_j(t)$ and summing over j from 1 to k , one has

$$\frac{1}{2} \frac{d}{dt} \|u^k\|_{H_0^1}^2 + \int_{\Omega} |\nabla u^k|^{p(x)} dx = \int_{\Omega} |u^k|^q \log |u^k| dx. \quad (4.11)$$

On the other hand, by multiplying (4.9) by $g_j'(t)$, sum over j from 1 to k and integrating with respect to time from 0 to t , we obtain

$$\int_0^t \|u_\tau^k(\tau)\|_{H_0^1}^2 d\tau + J(u^k) = J(u^k(0)). \quad (4.12)$$

We next distinguish two cases:

Case 1 $q < 2$. In this case, it is notice that $p^+ < q < 2$ and hence $X_0 = H_0^1(\Omega)$. Since $q < 2$, we can choose ϵ small enough such that $q + \epsilon < 2$ and estimate for $\int_{\Omega} |u^k|^q \log |u^k| dx$ as follows:

$$\int_{\Omega} |u^k|^q \log |u^k| dx \leq \frac{1}{e} \int_{\Omega} |u^k|^{q+\epsilon} dx = \frac{1}{e} \|u^k\|_{q+\epsilon}^{q+\epsilon} \leq C_1 \|u^k\|_2^{q+\epsilon} \leq C_1 \|u^k\|_{H_0^1}^{q+\epsilon}. \quad (4.13)$$

Substituting (4.13) into (4.11), we get

$$\frac{1}{2} \frac{d}{dt} \|u^k\|_{H_0^1}^2 \leq C_1 \|u^k\|_{H_0^1}^{q+\epsilon}.$$

Integrating the above inequality with respect to time from 0 to t and noticing that (4.10), we have

$$\begin{aligned} \|u^k\|_{H_0^1}^2 &\leq \|u^k(0)\|_{H_0^1}^2 + 2C_1 \int_0^t \|u^k(\tau)\|_2^{q+\epsilon} d\tau \\ &\leq C_2 + 2C_1 \int_0^t \|u^k(\tau)\|_2^{q+\epsilon} d\tau. \end{aligned}$$

Then since $\frac{q+\epsilon}{2} < 1$, by using Gronwall–Bellman–Bihari inequality, it follows from the above inequality that, for all $t \in [0, T]$,

$$\begin{aligned} \|u^k\|_{H_0^1}^2 &\leq \left(C_2^{\frac{2-q-\epsilon}{2}} + \frac{2-q-\epsilon}{2} C_1 t \right)^{\frac{2}{2-q-\epsilon}} \\ &\leq \left(C_2^{\frac{2-q-\epsilon}{2}} + \frac{2-q-\epsilon}{2} C_1 T \right)^{\frac{2}{2-q-\epsilon}} := C_3(T). \end{aligned} \quad (4.14)$$

It follows from (4.13) and (4.14) that

$$\begin{aligned} J(u^k) &= \int_{\Omega} \frac{1}{p(x)} |\nabla u^k|^{p(x)} dx - \frac{1}{q} \int_{\Omega} |u^k|^q \log |u^k| dx + \frac{1}{q^2} \|u^k\|_q^q \\ &\geq -\frac{1}{q} C_3^{\frac{q+\epsilon}{2}}. \end{aligned} \quad (4.15)$$

On the other hand, we deduce from (4.10) that $J(u^k(0)) \rightarrow J(u_0)$ as $k \rightarrow \infty$, and thus

$$J(u^k(0)) \leq C_4. \quad (4.16)$$

It follows from (4.12), (4.15) and (4.16) that

$$\int_0^t \|u_{\tau}^k(\tau)\|_{H_0^1}^2 d\tau \leq C_5. \quad (4.17)$$

Then from (4.14) and (4.17), for any $T > 0$, that there exists a u and a subsequence of $\{u^k\}_{k=1}^{\infty}$ (still denote by $\{u^k\}_{k=1}^{\infty}$) such that

$$\begin{aligned} u^k &\rightarrow u \quad \text{weakly star in } L^{\infty}(0, T; H_0^1(\Omega)), \\ u_t^k &\rightarrow u_t \quad \text{weakly in } L^2(0, T; L^2(\Omega)), \\ |\nabla u^k|^{p(x)-2} \nabla u^k &\rightarrow |\nabla u|^{p(x)-2} \nabla u \quad \text{weakly star in } L^{\infty}\left(0, T; L^{\frac{p(\cdot)}{p(\cdot)-1}}(\Omega)\right). \end{aligned}$$

Passing to the limit in (4.9) as $k \rightarrow \infty$, we have

$$\langle u_t, w_j \rangle + \langle \nabla u_t, \nabla w_j \rangle + \langle |\nabla u|^{p(x)-2} \nabla u, \nabla w_j \rangle = \langle |u|^{q-2} u \log |u|, w_j \rangle.$$

for $j = 1, 2, \dots, k$. Then since $\{w_j\}_{j=1}^{\infty}$ is a basis of X_0 , we have, for any $v \in X_0$,

$$\langle u_t, v \rangle + \langle \nabla u_t, \nabla v \rangle + \langle |\nabla u|^{p(x)-2} \nabla u, \nabla v \rangle = \langle |u|^{q-2} u \log |u|, v \rangle.$$

On the other hand, we obtain $u(0) = u_0$ due to (4.10). Thus u is a global solution of problem (1.1).

Case 2 $u_0 \in \mathcal{W}$. Since $u_0 \in \mathcal{W}$ and (4.10), we have, for sufficiently large k , $u^k(0) \in \mathcal{W}$. Then we prove that $u^k(t) \in \mathcal{W}$. In fact, by contradiction we assume that there exists a $t_0 > 0$ such that $u^k(t_0) \in \partial\mathcal{W}$. Noticing that 0 is an interior point of $\mathcal{W}$, we thus have $u^k(t_0) \neq 0$ and

$$I(u^k(t_0)) = 0 \quad \text{or} \quad J(u^k(t_0)) = d.$$

Since (4.12) and $u^k(0) \in \mathcal{W}$, we get $J(u^k(t_0)) \leq J(u^k(0)) < d$ and hence $J(u^k(t_0)) = d$ is not the case. Thus $I(u^k(t_0)) = 0$, $u^k(t_0) \neq 0$, i.e., $u^k(t_0) \in \mathcal{N}$. Then by the definition of d , we have $J(u^k(t_0)) \geq d$. This contradicts $J(u^k(t_0)) < d$. So, we get $u^k(t) \in \mathcal{W}$. Then we have

$$\frac{d}{dt} \|u^k\|_{H_0^1}^2 = -2I(u^k) \leq 0,$$

which, together with (4.10) implies that

$$\|u^k\|_{H_0^1}^2 \leq \|u^k(0)\|_{H_0^1}^2 \leq C_6. \quad (4.18)$$

On the other hand, since $u^k \in \mathcal{W}$, we have

$$\begin{aligned} d > J(u^k) &= \int_{\Omega} \left(\frac{1}{p(x)} - \frac{1}{q} \right) |\nabla u^k|^{p(x)} dx + \frac{1}{q^2} \|u^k\|_q^q + \frac{1}{q} I(u^k) \\ &\geq \left(\frac{1}{p^+} - \frac{1}{q} \right) \int_{\Omega} |\nabla u^k|^{p(x)} dx \geq 0, \end{aligned}$$

which implies that

$$\int_{\Omega} |\nabla u^k|^{p(x)} dx < \frac{p^+ q d}{q - p^+}. \quad (4.19)$$

It follows from (4.18) and (4.19) that

$$\{u^k\} \text{ is bounded in } L^\infty(0, \infty, X_0). \quad (4.20)$$

It follows from (4.12), $J(u^k) \geq 0$ and $J(u^k(0)) < d$ that

$$\{u_t^k\} \text{ is bounded in } L^2(0, \infty, H_0^1(\Omega)). \quad (4.21)$$

Then by analogous arguments in the proof of case 1, we deduce from (4.20) and (4.21) that problem (1.1) has a global solution. The proof is complete. $\square$

Next we state the asymptotic behavior for large time of the weak solutions.

Theorem 4.2 *Let (1.2)–(1.3) hold and $q < 2$. If $u_0 \in \mathcal{U}$ then the global weak solution u of problem (1.1) blows up at ∞ which means that $\lim_{t \rightarrow \infty} \|u(t)\|_{H_0^1}^2 = \infty$. Moreover,*

$$\frac{A_\eta}{t} (t - t_\eta)^{\frac{2}{2-q\eta}} \leq \|u(t)\|_{H_0^1}^2 \leq \left(B_\eta t + \|u_0\|_{H_0^1}^{(2-q)\eta} \right)^{\frac{2}{(2-q)\eta}},$$

for all $t \geq t_\eta$, where $\eta \in (0, 1)$ is arbitrary,

$$t_\eta = \frac{(1 + \eta) \|u_0\|_{H_0^1}^2}{2(1 - \eta)q(d - J(u_0))} > 0, \quad (4.22)$$

and the positive constants A_η and B_η are defined by

$$A_\eta = \left(1 - \frac{q\eta}{2}\right)^{\frac{2}{2-q\eta}} t_\eta^{-\frac{q\eta}{2-q\eta}} \left(\|u_0\|_{H_0^1}^2 + 2q(d - J(u_0))t_\eta\right) > 0, \quad (4.23)$$

$$B_\eta = \frac{|\Omega|^{\frac{(2-q)\eta}{2}} \eta}{e(1-\eta)} > 0. \quad (4.24)$$

Proof We first show the upper bound of the growth rate for the global weak solution of problem (1.1). Let $f(t)$ be the function defined by

$$f(t) = \|u(t)\|_{H_0^1}^2, \quad \text{for all } t \geq 0.$$

Since $f'(t) = -2I(u(t)) > 0$, we imply that

$$f(t) \geq f(0) = \|u_0\|_{H_0^1}^2 > 0, \quad \text{for all } t \geq 0.$$

By the definition of I and using Lemma 2.2 and Hölder inequality, one has

$$\begin{aligned} f'(t) &= -2 \int_{\Omega} |\nabla u(t)|^{p(x)} dx + 2 \int_{\Omega} |u(t)|^q \log |u(t)| dx \\ &\leq 2 \int_{\Omega} |u(t)|^q \frac{|u(t)|^{(2-q)(1-\eta)}}{e(2-q)(1-\eta)} dx \\ &= \frac{2}{e(2-q)(1-\eta)} \int_{\Omega} |u(t)|^{2-(2-q)\eta} dx \\ &\leq \frac{2|\Omega|^{\frac{(2-q)\eta}{2}}}{e(2-q)(1-\eta)} \|u(t)\|_2^{2-(2-q)\eta} \\ &\leq \frac{2|\Omega|^{\frac{(2-q)\eta}{2}}}{e(2-q)(1-\eta)} f^{\frac{2-(2-q)\eta}{2}}(t). \end{aligned}$$

From the last inequality and notice that $\frac{2-(2-q)\eta}{2} < 1$, we obtain

$$f(t) \leq \left(B_\eta t + f^{\frac{2}{(2-q)\eta}}(0)\right)^{\frac{2}{(2-q)\eta}},$$

i.e.,

$$\|u(t)\|_{H_0^1}^2 \leq \left(B_\eta t + \|u_0\|_{H_0^1}^{(2-q)\eta}\right)^{\frac{2}{(2-q)\eta}},$$

where B_η is defined in (4.24).

We next show the lower bound of the growth rate for the global weak solution of problem (1.1). Let us use again the function $F(t)$ introduced in the proof of Theorem 3.5. We deduce from (3.6) and (3.7) that

$$F(t)F''(t) \geq 2q \int_0^t \|u'(s)\|_{H_0^1}^2 ds \int_0^t \|u(s)\|_{H_0^1}^2 ds. \quad (4.25)$$

On the other hand, by Cauchy–Schwarz inequality, we obtain

$$\begin{aligned} \int_0^t \|u'(s)\|_{H_0^1}^2 ds \int_0^t \|u(s)\|_{H_0^1}^2 ds &\geq \left(\int_0^t \left[\int_{\Omega} u'(s)u(s) dx + \int_{\Omega} \nabla u'(s) \nabla u(s) dx \right] ds \right)^2 \\ &= \frac{1}{4} (F'(t) - F'(0))^2. \end{aligned} \quad (4.26)$$

Combining (4.25)–(4.26), one has

$$\begin{aligned} F(t)F''(t) &\geq \frac{q}{2} (F'(t) - F'(0))^2 \\ &\geq \frac{q}{2} (F'(t))^2 - qF'(t)F'(0). \end{aligned} \quad (4.27)$$

Take any $\eta \in (0, 1)$. It follows from (4.27) and (3.8) that

$$\begin{aligned} F(t)F''(t) - \frac{q\eta}{2} (F'(t))^2 &\geq \frac{q}{2} (1 - \eta) (F'(t))^2 - qF'(t)F'(0) \\ &= qF'(t) \left[\frac{1 - \eta}{2} F'(t) - \|u_0\|_{H_0^1}^2 \right] \\ &\geq qF'(t) \left[\frac{1 - \eta}{2} (\|u_0\|_{H_0^1}^2 + 2q(d - J(u_0))t) - \|u_0\|_{H_0^1}^2 \right]. \end{aligned}$$

This implies that

$$F(t)F''(t) - \frac{q\eta}{2} (F'(t))^2 \geq 0, \quad \text{for all } t \in [t_\eta, \infty),$$

where t_η is defined in (4.22). Setting $H(t) = F^{1-\frac{q\eta}{2}}(t)$, for $t > 0$, we get

$$H'(t) = \left(1 - \frac{q\eta}{2}\right) \frac{F'(t)}{F^{\frac{q\eta}{2}}(t)}, \quad \text{and} \quad H''(t) = \left(1 - \frac{q\eta}{2}\right) \frac{F(t)F''(t) - \frac{q\eta}{2} (F'(t))^2}{F^{1+\frac{q\eta}{2}}(t)}.$$

Since $q < 2$ and $\eta \in (0, 1)$, we have $\frac{q\eta}{2} < 1$, which gives $H''(t) \geq 0$, for all $t \in [t_\eta, \infty)$. Hence,

$$H(t) \geq H(t_\eta) + H'(t_\eta)(t - t_\eta), \quad \text{for all } t \in [t_\eta, \infty).$$

From this and $H(t_\eta) \geq 0$, we imply that

$$\begin{aligned} F(t) &= H^{\frac{2}{2-q\eta}}(t) \geq [H(t_\eta) + H'(t_\eta)(t - t_\eta)]^{\frac{2}{2-q\eta}} \\ &\geq (H'(t_\eta))^{\frac{2}{2-q\eta}} (t - t_\eta)^{\frac{2}{2-q\eta}}. \end{aligned} \quad (4.28)$$

We next estimate for $H'(t_\eta)$. Since $F''(t) = -2I(u(t)) > 0$, for all $t \geq 0$, one has

$$F(t) = \int_0^t F'(s) \, ds \leq tF'(t). \quad (4.29)$$

From this and (3.8), we imply that

$$\begin{aligned} H'(t_\eta) &= \left(1 - \frac{q\eta}{2}\right) \frac{F'(t_\eta)}{F^{\frac{q\eta}{2}}(t_\eta)} \\ &\geq \left(1 - \frac{q\eta}{2}\right) t_\eta^{-\frac{q\eta}{2}} (F'(t_\eta))^{1-\frac{q\eta}{2}} \\ &\geq \left(1 - \frac{q\eta}{2}\right) t_\eta^{-\frac{q\eta}{2}} \left(\|u_0\|_{H_0^1}^2 + 2q(d - J(u_0))t_\eta\right)^{1-\frac{q\eta}{2}}. \end{aligned} \quad (4.30)$$

It follows from (4.28) and (4.30) that

$$F(t) \geq A_\eta (t - t_\eta)^{\frac{2}{2-q\eta}}, \quad (4.31)$$

where A_η is defined in (4.23).

Combining (4.29) and (4.31), one has

$$\|u(t)\|_{H_0^1}^2 = F'(t) \geq \frac{F(t)}{t} \geq (t - t_\eta)^{\frac{2}{2-q\eta}} \frac{A_\eta}{t}, \quad \text{for all } t \geq t_\eta.$$

Taking this fact into account and notice that $\frac{2}{2-q\eta} > 1$, we imply that $\lim_{t \rightarrow \infty} \|u(t)\|_{H_0^1}^2 = \infty$. The proof is complete. $\square$

Theorem 4.3 *Let (1.2)–(1.3) hold. Assume that u is a global weak solutions to problem (1.1) with respect to the initial function $u_0 \in \mathcal{W}$. Then u decays at infinity, that is,*

$$\lim_{t \rightarrow \infty} \|\nabla u(t)\|_{p(\cdot)} = \lim_{t \rightarrow \infty} J(u(t)) = 0.$$

Moreover, if $p^- \geq 2$ then the following estimates hold true

(i) *If $p^+ = p^- = 2$ then*

$$\begin{aligned} \|\nabla u(t)\|_{p(\cdot)} &\leq \beta \left(J(u_0) + \|u_0\|_{H_0^1}^2 \right)^{\frac{1}{2}} e^{-\frac{\alpha}{2}t}, \\ J(u(t)) + \|u(t)\|_{H_0^1}^2 &\leq \left(J(u_0) + \|u_0\|_{H_0^1}^2 \right) e^{-\alpha t}, \end{aligned}$$

for all $t > 0$. Here α and β are some positive constants.

(ii) *If $p^+ > 2$ then*

$$\|\nabla u(t)\|_{p(\cdot)} \leq \zeta \left[\frac{p^+ - 2}{2} \gamma t + \left(J(u_0) + \|u_0\|_{H_0^1}^2 \right)^{\frac{2-p^+}{2}} \right]^{-\frac{2}{p^+(p^+-2)}},$$

$$J(u(t)) + \|u(t)\|_{H_0^1}^2 \leq \left[\frac{p^+ - 2}{2} \gamma t + \left(J(u_0) + \|u_0\|_{H_0^1}^2 \right)^{\frac{2-p^+}{2}} \right]^{-\frac{2}{p^+-2}},$$

for all $t > 0$. Here γ and ζ are some positive constants.

Proof We first prove the decay property of global weak solutions to problem (1.1). It is first noticed that $u(t) \in \mathcal{W}$ since $u_0 \in \mathcal{W}$. Then by virtue of Lemma 2.1, there exists $s_* > 1$ such that $I(s_* u(t)) = 0$. Hence, by analogous arguments in the proof of Lemma 2.1, we have

$$\begin{aligned} s_*^q I(u(t)) &= \int_{\Omega} (s_*^q - s_*^{p(x)}) |\nabla u(t)|^{p(x)} dx + s_*^q \log s_* \|u(t)\|_q^q \\ &\geq (s_*^q - s_*^{p^+}) \int_{\Omega} |\nabla u(t)|^{p(x)} dx + s_*^q \log s_* \|u(t)\|_q^q. \end{aligned}$$

Dividing the above inequality by s_*^q , we get

$$I(u(t)) \geq (1 - s_*^{p^+-q}) \int_{\Omega} |\nabla u(t)|^{p(x)} dx + \log s_* \|u(t)\|_q^q. \quad (4.32)$$

As the next step, we estimate for s_* . Replacing u by $s_* u(t)$ in (2.2), then by definition d , one has

$$\begin{aligned} d \leq J(s_* u(t)) &= \int_{\Omega} s_*^{p(x)} \left(\frac{1}{p(x)} - \frac{1}{q} \right) |\nabla u(t)|^{p(x)} dx + \frac{s_*^q}{q^2} \|u(t)\|_q^q \\ &\leq s_*^q \left[\int_{\Omega} \left(\frac{1}{p(x)} - \frac{1}{q} \right) |\nabla u(t)|^{p(x)} dx + \frac{1}{q^2} \|u(t)\|_q^q \right] \\ &= s_*^q \left[J(u(t)) - \frac{1}{q} I(u(t)) \right] \\ &\leq s_*^q J(u_0), \end{aligned}$$

which, together with $J(u_0) < d$, implies that

$$s_* \geq \left(\frac{d}{J(u_0)} \right)^{\frac{1}{q}} > 1. \quad (4.33)$$

It follows from (4.32) and (4.33) that

$$\begin{aligned}
I(u(t)) &\geq \left(1 - \left(\frac{d}{J(u_0)}\right)^{\frac{p^+-q}{q}}\right) \int_{\Omega} |\nabla u(t)|^{p(x)} dx + \frac{1}{q} \log \left(\frac{d}{J(u_0)}\right) \|u(t)\|_q^q \\
&\geq C_1 \left(\int_{\Omega} |\nabla u(t)|^{p(x)} dx + \|u(t)\|_q^q \right),
\end{aligned}$$

where

$$C_1 = \min \left\{ 1 - \left(\frac{d}{J(u_0)}\right)^{\frac{p^+-q}{q}}, \frac{1}{q} \log \left(\frac{d}{J(u_0)}\right) \right\} > 0.$$

Then it follows that

$$\begin{aligned}
J(u(t)) &= \int_{\Omega} \left(\frac{1}{p(x)} - \frac{1}{q}\right) |\nabla u(t)|^{p(x)} dx + \frac{1}{q^2} \|u(t)\|_q^q + \frac{1}{q} I(u(t)) \\
&\leq \left(\frac{1}{p^-} - \frac{1}{q}\right) \int_{\Omega} |\nabla u(t)|^{p(x)} dx + \frac{1}{q^2} \|u(t)\|_q^q + \frac{1}{q} I(u(t)) \\
&\leq C_2 I(u(t)),
\end{aligned} \tag{4.34}$$

where

$$C_2 = \frac{1}{C_1} \max \left\{ \frac{1}{p^-} - \frac{1}{q}, \frac{1}{q^2} \right\} + \frac{1}{q} > 0.$$

By approximation, without loss of generality, we can assume that u be a sufficiently smooth solution of problem (1.1). Take $v = u_t$ in Definition 3.1, we obtain

$$\frac{d}{dt} J(u(t)) = -\|u'(t)\|_{H_0^1}^2. \tag{4.35}$$

Next, we define an auxiliary function $L : [0, \infty) \rightarrow (0, \infty)$ by

$$L(t) = J(u(t)) + \|u(t)\|_{H_0^1}^2. \tag{4.36}$$

It follows from (4.34), (4.35) and (4.36) that

$$L'(t) = -\|u'(t)\|_{H_0^1}^2 - 2I(u(t)) \leq -2I(u(t)) \leq -\frac{2}{C_2} J(u(t)) \leq 0. \tag{4.37}$$

Thus $L(t)$ is non-increasing with respect to t . From this and $L(t) \geq 0$, there exist a constant c such that $\lim_{t \rightarrow \infty} L(t) = c$. Hence

$$\int_0^\infty L'(t) dt = c - L(0) < \infty.$$

Therefore there exists a sequence $\{t_n\}$ with $t_n \rightarrow \infty$ as $n \rightarrow \infty$ such that

$$\lim_{n \rightarrow \infty} L'(t_n) = 0,$$

which, together with (4.37), implies that

$$\lim_{n \rightarrow \infty} J(u(t_n)) = 0.$$

It is from this and $J(u(t))$ is non-increasing with respect to t that

$$\lim_{t \rightarrow \infty} J(u(t)) = 0. \quad (4.38)$$

On the other hand, since $u(t) \in \mathcal{W}$, we have

$$\begin{aligned} J(u(t)) &= \int_{\Omega} \left(\frac{1}{p(x)} - \frac{1}{q} \right) |\nabla u(t)|^{p(x)} dx + \frac{1}{q^2} \|u(t)\|_q^q + \frac{1}{q} I(u(t)) \\ &\geq \left(\frac{1}{p^+} - \frac{1}{q} \right) \min \left\{ \|\nabla u(t)\|_{p(\cdot)}^{p^-}, \|\nabla u(t)\|_{p(\cdot)}^{p^+} \right\}, \end{aligned}$$

which implies that

$$\|\nabla u(t)\|_{p(\cdot)} \leq \max \left\{ \left(\frac{qp^+ J(u(t))}{q - p^+} \right)^{\frac{1}{p^-}}, \left(\frac{qp^+ J(u(t))}{q - p^+} \right)^{\frac{1}{p^+}} \right\}. \quad (4.39)$$

This, together with (4.38), implies that

$$\lim_{t \rightarrow \infty} \|\nabla u(t)\|_{p(\cdot)} = 0.$$

We next show decay estimates of global weak solutions to problem (1.1) when $p^- \geq 2$. Since $J(u(t)) \leq J(u_0)$ and (4.39), one has

$$\|\nabla u(t)\|_{p(\cdot)} \leq \max \left\{ \left(\frac{qp^+ J(u_0)}{q - p^+} \right)^{\frac{1}{p^-}}, \left(\frac{qp^+ J(u_0)}{q - p^+} \right)^{\frac{1}{p^+}} \right\} := C_3.$$

Then

$$\begin{aligned} J(u(t)) &= \int_{\Omega} \left(\frac{1}{p(x)} - \frac{1}{q} \right) |\nabla u(t)|^{p(x)} dx + \frac{1}{q^2} \|u(t)\|_q^q + \frac{1}{q} I(u(t)) \\ &\geq \left(\frac{1}{p^+} - \frac{1}{q} \right) \min \left\{ \|\nabla u(t)\|_{p(\cdot)}^{p^-}, \|\nabla u(t)\|_{p(\cdot)}^{p^+} \right\} \\ &= \left(\frac{1}{p^+} - \frac{1}{q} \right) \min \left\{ \|\nabla u(t)\|_{p(\cdot)}^{p^- - p^+}, 1 \right\} \|\nabla u(t)\|_{p(\cdot)}^{p^+} \\ &\geq \left(\frac{1}{p^+} - \frac{1}{q} \right) \min \left\{ C_3^{p^- - p^+}, 1 \right\} \|\nabla u(t)\|_{p(\cdot)}^{p^+}. \end{aligned} \quad (4.40)$$

By using (4.40) and the embedding $W_0^{1,p(\cdot)}(\Omega) \hookrightarrow H_0^1(\Omega)$ for $p^- \geq 2$, we get

$$\|u(t)\|_{H_0^1}^2 \leq S_H^2 \|\nabla u(t)\|_{p(\cdot)}^2 \leq C_4 J_{\frac{2}{p^+}}(u(t)), \quad (4.41)$$

where S_H is defined in (3.2) and

$$C_4 = S_H^2 \left[\left(\frac{1}{p^+} - \frac{1}{q} \right) \min \left\{ C_3^{p^- - p^+}, 1 \right\} \right]^{-\frac{2}{p^+}}.$$

Then by (4.41) and (4.36), we obtain

$$L(t) \leq J(u(t)) + C_4 J_{\frac{2}{p^+}}(u(t)). \quad (4.42)$$

If $p^+ = p^- = 2$ then the above inequality can be rewritten as

$$L(t) \leq (1 + C_4) J(u(t)). \quad (4.43)$$

We deduce from (4.43) and (4.37) that

$$L'(t) \leq -\alpha L(t),$$

which yields that

$$L(t) \leq L(0)e^{-\alpha t},$$

where

$$\alpha = \frac{2}{C_2(1 + C_4)}.$$

This above inequality can be rewritten as

$$J(u(t)) + \|u(t)\|_{H_0^1}^2 \leq \left(J(u_0) + \|u_0\|_{H_0^1}^2 \right) e^{-\alpha t}. \quad (4.44)$$

Then by (4.40) and (4.44), we get

$$\|\nabla u(t)\|_{p(\cdot)} \leq \beta \left(J(u_0) + \|u_0\|_{H_0^1}^2 \right)^{\frac{1}{2}} e^{-\frac{\alpha}{2}t},$$

where

$$\beta = \left(\frac{1}{2} - \frac{1}{q} \right)^{-\frac{1}{2}}.$$

Otherwise, if $p^+ > 2$, then using again the inequality (4.42) once more and notice that $J(u(t)) \leq J(u_0) < d$, one has

$$\begin{aligned}
 L(t) &\leq \left[J^{\frac{p^+-2}{p^+}}(u(t)) + C_4 \right] J^{\frac{2}{p^+}}(u(t)) \\
 &\leq \left[d^{\frac{p^+-2}{p^+}} + C_4 \right] J^{\frac{2}{p^+}}(u(t)).
 \end{aligned} \tag{4.45}$$

We deduce from (4.45) and (4.37) that

$$L'(t) \leq -\gamma L^{\frac{p^+}{2}}(t),$$

where γ is a positive constant given by

$$\gamma = \frac{2}{C_2 \left[d^{\frac{p^+-2}{p^+}} + C_4 \right]^{\frac{p^+}{2}}}.$$

This implies that

$$L(t) \leq \left(\frac{p^+-2}{2} \gamma t + L^{\frac{2-p^+}{2}}(0) \right)^{-\frac{2}{p^+-2}},$$

which can be rewritten as

$$J(u(t)) + \|u(t)\|_{H_0^1}^2 \leq \left[\frac{p^+-2}{2} \gamma t + \left(J(u_0) + \|u_0\|_{H_0^1}^2 \right)^{\frac{2-p^+}{2}} \right]^{-\frac{2}{p^+-2}}. \tag{4.46}$$

Then by (4.40) and (4.46), we get

$$\|\nabla u(t)\|_{p(\cdot)} \leq \zeta \left[\frac{p^+-2}{2} \gamma t + \left(J(u_0) + \|u_0\|_{H_0^1}^2 \right)^{\frac{2-p^+}{2}} \right]^{-\frac{2}{p^+(p^+-2)}},$$

where

$$\zeta = \left[\left(\frac{1}{p^+} - \frac{1}{q} \right) \min \left\{ C_3^{p^--p^+}, 1 \right\} \right]^{-\frac{1}{p^+}}.$$

The proof is complete. $\square$

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